# Interoceptomics Simple Script

The essential spoken lines follow the same 111 pages and 135 reveals as the presentation. Advance through the cues within each page. The experiments and skill-transfer goals are proposals.

## I Know Kung Fu

Slide 1 | Reveals 1

I know kung fu. In the film, a file becomes a skill. Our first question is smaller: can recorded signals predict a movement we can measure? Helping a body move and teaching a lasting skill need different tests. The film cuts straight to the dojo. We have to account for everything in between.

### Reveal Cues

1. Show the title, image and subtitle together. Explain Neo and the BCI subtitle together; distinguish instructions, assistance and retained skill.

### References

The Matrix (1999), official film page https://www.warnerbros.com/movies/matrix

Brain2Body programme ethos, current source

## High-Fidelity Telepathy

Slide 2 | Reveals 2

Professor X makes it look easy. A real link needs something to record, a way to deliver it, and a recipient who can use it. High fidelity has to mean accurate for a named task. Useful information transfer would still leave thought reading and identical experience unproved.

### Reveal Cues

1. Show the title, image and subtitle together. Use Professor X to introduce the question; define task-specific fidelity and an observable response.

### References

Brain2Body programme ethos, current source

## It's Already Here

Slide 3 | Reveals 3

A hand draws a hand. Each mark changes what we see, and what we see can change the next mark. An unusual connection between two nervous systems asks a sharper question: what information crossed, and what could the recipient do with it? A matching response doesn't prove matching private experience.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the hand-drawing image and subtitle; explain action, sensation and the limits of response agreement.

### References

Brain2Body programme ethos, current source

## How They Think

Slide 4 | Reveals 4

These are Krista and Tatiana Hogan: two people with separate brains and a remarkable connection. Their documentary reports shared sight, taste and touch. That makes the case worth studying. It doesn't establish unrestricted thought reading, identical experiences or a laboratory design we can simply copy.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the twins photo and exact subtitle; name Krista and Tatiana and identify documentary-report limits.

### References

Curious Features, Inseparable documentary https://curiousfeatures.ca/project/inseparable/

## Share A Brain

Slide 5 | Reveals 5

Documentary accounts describe separate brains with a connection around the thalamus, and report shared sight, taste and touch. The CBC documentary image is an illustration, not a clinical scan; we have not verified their individual clinical anatomy. A corresponding experiment would need a known stimulus, a controlled route and a response scored independently.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the thalamic-bridge illustration and sensory subtitle; retain reported/schematic scope.

### References

Curious Features, Inseparable documentary https://curiousfeatures.ca/project/inseparable/

Brain2Body programme ethos, current source

Wellcome Collection, The twins who share a brain, Channel 4 Body Shock, broadcast 25 May 2010 https://wellcomecollection.org/works/qugqdhqy

## Want To Recreate This

Slide 6 | Reveals 6

We want a useful link, but first we need to know what its signals mean. A model can memorize a schedule or a recording session and look clever. Measure the actual movement, hold out genuinely new conditions, and compare with a simpler model. Then the prediction has somewhere to fail.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the lurking crocodile and subtitle; expose shortcut learning and the need for independent targets.

### References

Brain2Body programme ethos, current source

# Human Genome Project Failed

Slide 7 | Reveals 7

The sequencing project succeeded. The treatment problem remains enormous. About 300 million people live with rare diseases; the FDA says only about 5 percent of recognized rare conditions had an FDA-approved treatment as of 2025. Those figures do not establish that 300 million people receive no treatment, or no care. A map is a beginning. Effective intervention needs more.

### Reveal Cues

1. Show the title, image and subtitle together. Explain DNA image and subtitle; explain the 300 million people versus~5% conditions denominators.

### References

NHGRI, The Human Genome Project https://www.genome.gov/human-genome-project

FDA, A New Framework for Rare Disease Review, October 2026 https://www.fda.gov/media/195694/download?attachment=

WHO, World Health Assembly daily update, 24 May 2025 https://www.who.int/news/item/24-05-2025-seventy-eighth-world-health-assembly---daily-update--24-may-2025/

NIH NCATS, Rare Diseases Fact Sheet https://ncats.nih.gov/sites/default/files/NCATS_RareDiseasesFactSheet.pdf

# 2.5 Decades

Slide 8 | Reveals 8

Knock-in and knockout studies ask what changes when a specified genetic component changes. They can reveal a contribution, but compensation and context still matter. Two and a half decades is a rounded framing, not a universal biological timetable. The aim is faster understanding through better tests, with each result kept inside its evidence.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the rounded interval, controlled genetic perturbation and context.

### References

NHGRI, Knockout https://www.genome.gov/genetics-glossary/Knockout

Brain2Body programme ethos, current source

# Incremental Improvement

### 1 Gene At A Time

Slide 9 | Reveals 9

Changing one Lego brick is easy to imagine. A living system also depends on interactions, timing and context. Genetics already uses much broader experiments than one gene at a time. Our own extra sensors must earn their place the same way: better predictions on independent tests, rather than a bigger pile of inputs.

### Reveal Cues

1. Show Incremental Improvement, the image and 1 Gene At A Time together. Explain the one-gene example, interactions and modern multiplexed work.

### References

NHGRI, Gene Regulation https://www.genome.gov/genetics-glossary/Gene-Regulation

Brain2Body programme ethos, current source

## Genes ≠ Traits

Slide 10 | Reveals 10

A gene is not the same thing as a trait. Genes, development and environment can all contribute, and some genetic effects are well established. A neural map likewise constrains an explanation; a functional test has to show what happens in the task.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the gene–trait distinction and connect the analogy to an independently tested neural prediction.

### References

NHGRI, Phenotype https://www.genome.gov/genetics-glossary/Phenotype

Brain2Body programme ethos, current source

## Form ≠ Function

Slide 11 | Reveals 11

A Croc isn't a crocodile. Resemblance and labels don't establish what something does. Neural wiring constrains possible behavior; activity, body state and inputs still matter. Test a specific prediction against an independent observation. Structure gives us a better question. Function supplies the next test.

### Reveal Cues

1. Show the title, image and subtitle together. Explain Crocs and Function=Function; explain a functional test and the continuing value of anatomy.

### References

Brain2Body programme ethos, current source

## Disorder To Disease

Slide 12 | Reveals 12

The virtual Atlas robots show training in parallel simulations. AI can help find relationships in complex recordings. It doesn't turn a disorder into an understood disease by naming it. Choose a useful prediction, compare it fairly and test it under new conditions. Any clinical benefit needs evidence beyond success on a recorded movement task.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the parallel virtual humanoids, then distinguish data, a bounded prediction, deployment and evidence of clinical benefit.

### References

Brain2Body programme ethos, current source

# Dumb Medical Models

Slide 13 | Reveals 13-14

The goal is a model that makes useful, testable predictions with the relevant context. A complex model is not automatically better than a simple clinical rule. Compare them fairly and test any clinical benefit separately.

### Reveal Cues

1. Reveal Dumb Medical Models with its outgoing down arrow; frame the label as a rhetorical starting point.

2. Add Smart Medical Models; define useful performance through fair comparison, independent testing and limits.

### References

Brain2Body programme ethos, current source

# Deepblue To Deepmind

Slide 14 | Reveals 15

Deep Blue used high-speed search for chess. DeepMind's AlphaGo combined learned representations with search for Go. Both solved defined tasks; the comparison does not make deeper networks automatically general or biologically explanatory. Test the learned features on genuinely new conditions.

### Reveal Cues

1. Show the title, image and subtitle together. Distinguish IBM's Deep Blue from DeepMind's AlphaGo and retain the task-specific comparison.

### References

IBM, Deep Blue https://www.ibm.com/history/deep-blue

Silver et al. (2016), Mastering the game of Go with deep neural networks and tree search https://pubmed.ncbi.nlm.nih.gov/26819042/

Brain2Body programme ethos, current source

# Representation

Slide 15 | Reveals 16

Deep learning can combine features into a representation useful for a prediction. Test whether those features carry the intended information on new data; they may also capture artefacts. More layers alone do not prove a biological explanation.

### Reveal Cues

1. Show the title, image and subtitle together. Explain learned representation and the independent test needed to distinguish useful features from artefacts.

### References

Google DeepMind, AlphaGo https://deepmind.google/research/alphago/

Brain2Body programme ethos, current source

## IBM Watson To AlphaGo

Slide 16 | Reveals 17

Watson answered Jeopardy! questions; AlphaGo chose moves in Go. They used different methods on different tasks. The useful lesson is to define an input and an outcome that can be checked, not to treat the systems as a direct benchmark.

### Reveal Cues

1. Show IBM Watson To AlphaGo and its illustration. Identify the different tasks and connect them to a checkable prediction.

### References

IBM Research, Building Watson: An Overview of the DeepQA Project https://research.ibm.com/publications/building-watson-an-overview-of-the-deepqa-project

Google DeepMind, AlphaGo https://deepmind.google/research/alphago/

Brain2Body programme ethos, current source

## Autocomplete To Attention

Slide 17 | Reveals 18

Autocomplete predicts a continuation. Attention can relate information across a sequence, and a Transformer can also predict text. Context for our experiment still needs aligned sensors, body state and task conditions; generalisation needs tests on named new conditions.

### Reveal Cues

1. Show the title, image and subtitle together. Explain autocomplete and attention as compatible ideas, then qualify the ambition to generalise.

### References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

Brain2Body programme ethos, current source

## Context

Slide 18 | Reveals 19

Transformers use attention to relate information across a sequence. Context still needs a shared clock, task and body state. Test new people or conditions explicitly rather than assuming a long context makes a model general.

### Reveal Cues

1. Show the title, image and subtitle together. Explain attention as a way to use context, then identify the measured context and held-out test.

### References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

Brain2Body programme ethos, current source

## Google Translate To ChatGPT

Slide 19 | Reveals 20

Google Translate and ChatGPT illustrate different language interfaces, not a direct comparison on one task. Our proposed translation must link specified observations to an independently measured outcome; a convincing answer does not prove copied experience.

### Reveal Cues

1. Show Google Translate To ChatGPT and its illustration. Distinguish language products from a tested brain-body mapping.

### References

Google Research, A Neural Network for Machine Translation, at Production Scale https://research.google/blog/a-neural-network-for-machine-translation-at-production-scale/

OpenAI, Introducing ChatGPT https://openai.com/index/chatgpt/

Brain2Body programme ethos, current source

## GPT-3 To O3

Slide 20 | Reveals 21

GPT-3 and o3 name language models; an agent also needs a way to act and observe what happened. Its score must match the real objective. Our predictions answer to independent movement measurements; a simulator alone does not establish biological accuracy.

### Reveal Cues

1. Show the title, image and subtitle together. Distinguish model names from an agent workflow and identify an independent outcome.

### References

OpenAI, Language Models Are Few-Shot Learners https://openai.com/index/language-models-are-few-shot-learners/

OpenAI, o3 model documentation https://developers.openai.com/api/docs/models/o3

Yao et al. (2023), ReAct: Synergizing Reasoning and Acting in Language Models https://arxiv.org/abs/2210.03629

Brain2Body programme ethos, current source

## Grounding

Slide 21 | Reveals 22

Reinforcement can use feedback from a check, but the check may miss an important error. Check the prediction against an independent outcome that could prove it wrong. Use the same target and information for a simpler comparison. A simulator's approval is not the same as biological or clinical validation.

### Reveal Cues

1. Show the title, image and subtitle together. Explain how a check can inform reinforcement while a calibrated independent outcome grounds the claim.

### References

Brain2Body programme ethos, current source

Helff et al. (2026), LLMs Gaming Verifiers: RLVR can Lead to Reward Hacking https://arxiv.org/abs/2604.15149

## Task Augmentation To Automation

Slide 22 | Reveals 23

Augmentation helps a person perform a task; automation carries out a specified part of it. Record who does what and measure the outcome. Help during practice is different from a skill retained after guidance stops.

The images show collaborative sawing and an industrial robot production line. They illustrate the contrast; they do not establish the proposed system.

### Reveal Cues

1. Show the title and both side-by-side photographs. Distinguish the collaborative sawing experiment from industrial automation and from the proposed system.

### References

Brain2Body programme ethos, current source

Luka Peternel, human–robot collaborative sawing, CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Human-Robot-Collaboration-Sawing-2016-Luka-Peternel.jpg

Clemenspool, industrial robot production line, CC0 https://commons.wikimedia.org/wiki/File:Staeubli_Robots_line.jpg

## REPRESENTATION

Slide 23 | Reveals 24-26

Representation: what information did we keep? Context: which body, task and moment does it belong to? Grounding: what independent observation can show it is wrong? Each arrow adds a requirement. Together they give us a prediction we can test, rather than three impressive labels.

### Reveal Cues

1. Reveal REPRESENTATION with its following arrow; discuss the information retained.

2. Add CONTEXT with its following arrow; preserve the first item and connect task/time conditions.

3. Add GROUNDING; identify independently measured movement and its limits.

### References

Brain2Body programme ethos, current source

## For End-To-End Emulation

Slide 24 | Reveals 27

A loop closes when action changes the next observation and that observation can affect the next action. Offline prediction, simulation and a living body's feedback are different cases. Specify the route and its timing, then demonstrate the return. Arrows alone don't establish an end-to-end biological loop.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

## DNA

Slide 25 | Reveals 28-30

DNA can be transcribed into RNA, and messenger RNA can be translated into protein. Regulation and cell state shape every step; some RNAs aren't translated at all. This chain opens the question rather than closing it: which relationships can we predict, and which consequences have we actually measured?

### Reveal Cues

1. Show DNA and its down arrow; distinguish sequence from activity.
2. Add RNA and its down arrow; explain expression, processing and noncoding scope.
3. Add PROTEIN; explain why structure and living consequence still require tests.

### References

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

## AlphaGenome

Slide 26 | Reveals 31-33

AlphaGenome predicts regulatory measurements from DNA. AlphaFold predicts molecular structures. AlphaTranscriptome is our proposed missing component, not a released model shown here. Single-cell models provide useful precedents. Connecting these names still requires compatible inputs, independent tests and evidence of the actual loop.

### Reveal Cues

1. Reveal AlphaGenome with its arrow; explain its DNA input and genomic targets.
2. Add AlphaTranscriptome with its arrow; explicitly identify the intended AlphaTranscriptome as proposed.
3. Add AlphaFold; explain measured structure benchmarks and the unverified integration between components.

### References

Avsec et al. (2026), Advancing regulatory variant effect prediction with AlphaGenome https://pmc.ncbi.nlm.nih.gov/articles/PMC12851941/

Cui et al. (2024), scGPT: toward building a foundation model for single-cell multi-omics using generative AI https://www.nature.com/articles/s41592-024-02201-0

Jumper et al. (2021), Highly accurate protein structure prediction with AlphaFold https://pubmed.ncbi.nlm.nih.gov/34265844/

## An End-To-End Pipeline

Slide 27 | Reveals 34

An end-to-end pipeline needs compatible inputs, transformations and a measured output. AlphaTranscriptome is proposed; the named components are not demonstrated here as one integrated system. Running a pipeline also does not by itself close a feedback loop.

### Reveal Cues

1. Show the title, image and subtitle together. Describe the proposed molecular pipeline and distinguish end-to-end processing from demonstrated closed-loop emulation.

### References

Brain2Body programme ethos, current source

## A Closed Loop

Slide 28 | Reveals 35

For neurology, start with an outcome we can check: movement. Connect structure, activity, muscles, mechanics and sensory consequences around that same task. An initial prediction is narrower than a working biological loop. State the endpoints, compare with a simpler model, and keep the measured errors visible.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## Start With Movement

Slide 29 | Reveals 36

Movement gives us consequences we can test. The pool meme highlights a safety problem, not a claim about people who cannot move their limbs. Our experiment begins with a safe, defined task and independent measures of timing and success. The stakes motivate the question; they don't determine the procedure.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

## Movement Is The Ultimate Problem

Slide 30 | Reveals 37

Movement turns many biological demands into something we can measure. A body reaches, avoids or approaches. That gives a prediction an external target. It doesn't make every brain function movement, and the same trajectory can have different intentions. It gives us a firm place to begin testing.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

## Whole-Brain Emulation

Slide 31 | Reveals 38

A brain acts through a body and receives consequences through that body. Model the relationships relevant to the task, including local spinal processing. Then test whether body measurements improve a prediction beyond brain measurements alone. A complete-looking anatomy is still different from a validated account of function.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

## Virtual Molecules

Slide 32 | Reveals 39-40

Molecules and cells occupy different physical scales. Model detail and statistical averaging are separate choices. To connect them, state the quantities, units and conditions crossing the interface. A detailed molecular picture doesn't automatically explain a cell; an average cell result doesn't verify every molecule.

### Reveal Cues

1. Reveal Virtual Molecules with its following down arrow; define a physical scale and a target.

2. Add Virtual Cells; explain the proposed interface and independent tests in each direction.

### References

Brain2Body programme ethos, current source

## Virtual Bodies

Slide 33 | Reveals 41-42

A virtual brain beside a virtual body isn't yet a working connection. Define the control signals, mechanics, feedback and shared clock. Test predictions in both directions, keeping ambiguous hidden states uncertain. A simulator helps express the question; independent measurements decide how well the coupled model answers it.

### Reveal Cues

1. Show Virtual Bodies with its following down arrow; identify mechanical and measurement assumptions.

2. Add Virtual Brains; explain control/feedback interfaces and separately tested directions.

### References

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

Brain2Body programme ethos, current source

## Brains REPRESENT Environments.

Slide 34 | Reveals 43-45

A brain can carry information useful for acting in its surroundings. That isn't a little picture of the room inside the head. Define what the representation should predict, then test it under changed conditions. A successful decoder shows useful information was available; it doesn't reveal a unique complete neural code.

The same reach starts from a particular body state. Pose, load and fatigue can change what a signal means. Measure useful context and test whether it improves a reserved prediction. A trajectory can still have several hidden explanations. More context narrows uncertainty; it doesn't automatically eliminate it.

The world gives a prediction somewhere to be wrong. Check the actual action, timing and consequences. Record which information the model received; hidden shortcuts change the task. Simulated success and measured biological agreement need separate tests. The observable outcome constrains the explanation without revealing a unique private experience.

### Reveal Cues

1. Reveal Brains REPRESENT Environments with its outgoing down arrow; define a testable prediction.

2. Add Bodies CONTEXTualise Brains with its outgoing down arrow; identify the relevant body state.

3. Add Environments GROUND Brains; identify an independent measured outcome.

### References

Wolpert, Ghahramani and Jordan (1995), An internal model for sensorimotor integration
https://pubmed.ncbi.nlm.nih.gov/7569931/

Brain2Body programme ethos, current source

## REPRESENTATION

Slide 35 | Reveals 46

Keep all three together: what information is represented, which situation gives it context, and what independent outcome can show the prediction is wrong? Each part needs its own test.

### Reveal Cues

1. Show REPRESENTATION, CONTEXT, GROUNDING and both arrows together immediately; connect them to the three preceding statements.

### References

Brain2Body programme ethos, current source

## Kung Fu THE HARD WAY

Slide 36 | Reveals 47

The hard way keeps the missing steps: receive information, act, encounter consequences and test what changed. Measurement, prediction, assistance and lasting learning are different achievements. The roadmap connects them without claiming they're complete. Each step needs an outcome that can show it failed.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

### References

Brain2Body programme ethos, current source

## 1. Engineer The World's Greatest Model Organisms Platform

Slide 37 | Reveals 48

Build a useful organism platform and test its observations. “Greatest” is the ambition, not a demonstrated comparison.

### Reveal Cues

1. Show numbered item 1; explain the platform and unmeasured superlative.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## 2. Turn That Organism Human

Slide 38 | Reveals 49

Study a specified human-related feature; this does not make the whole organism human.

### Reveal Cues

1. Show numbered item 2; narrow humanisation to specified features.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## 3. Build The Blood Pressure Monitor For The Nervous System

Slide 39 | Reveals 50

Measure a declared nervous-system quantity against calibrated references. A universal nervous-system equivalent of blood pressure is not established here.

### Reveal Cues

1. Show numbered item 3; explain calibrated measurement and the absent universal nervous-system number.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## 4. Build The Pacemaker For The Nervous System

Slide 40 | Reveals 51

The pacemaker analogy asks about a separately qualified output interface and its measured consequence; it is not an established treatment.

### Reveal Cues

1. Show numbered item 4; distinguish delivery from recording and retain separate interface qualification.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## 5. Use It To Make You Dance

Slide 41 | Reveals 52

Measure the action, then distinguish guided assistance from a skill retained when trainer-specific help ends.

### Reveal Cues

1. Show numbered item 5; distinguish assistance, retention and transfer.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## STAGE 0: Brain2Body

Slide 42 | Reveals 53

Stage zero asks how declared neural and body observations relate in a bounded task. It is a proposed baseline, not a complete translation.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## 0.1: NeuroTranslate

Slide 43 | Reveals 54

Start with a defined brain-body mapping and test its predictions against measured movement. The artificial womb is a proposal; the newborn image is a different-species proxy, not a verified Antechinus model.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## 0.2: Navigation

Slide 44 | Reveals 55

Start with a straight-walled maze. Measure where the animal goes and when, then test the model against that outcome. The maze and animal are illustrations of a proposed experiment.

The proposed LiDAR illustration places translucent red above the maze floor and blue below it. It does not establish a working measurement system.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## 0.3: NeuroRelay

Slide 45 | Reveals 56

Send a declared signal through the proposed NeuroRelay, then test whether the recipient uses information that a control link does not provide. This is a proposed experiment.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## 0.4: Orientation

Slide 46 | Reveals 57

Now use a winding maze. Test whether the mapping still works when direction and route change, using measured outcomes and conditions the model has not already seen.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## 0.5: NeuroRelay

Slide 47 | Reveals 58

Test NeuroRelay again in the turning task. The first route cannot certify the second; the proposed link needs a separate comparison and measured outcome.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

## STAGE 1: Geno-Humanisation

Slide 48 | Reveals 59

Geno-humanisation asks whether a specified genetic feature can make a model more informative. It does not make the whole animal human.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

### References

Brain2Body canonical glossary, revision 33

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

Jinek et al. (2012), A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity https://pubmed.ncbi.nlm.nih.gov/22745249/

## 1.1: Geno-Transplantation

Slide 49 | Reveals 60

Test a specific human-related genetic feature, not the idea of making the whole animal human. The illustrated needle is a proposal; the genetic change and its effects are not established.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Brain2Body canonical glossary, revision 33

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

Jinek et al. (2012), A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity https://pubmed.ncbi.nlm.nih.gov/22745249/

# STAGE 2: Xeno-Humanisation

Slide 50 | Reveals 61

Xeno-humanisation asks what specified donor tissue contributes in a host. It is a feature-specific proposal requiring independent evidence.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

### References

Revah et al. (2022; updated article with 2026 Author Correction), Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-022-05277-w

Brain2Body canonical glossary, revision 33

Brain2Body programme ethos, current source

# 2.1: Xeno-Transplantation

Slide 51 | Reveals 62

Ask what donor tissue survives, connects and contributes. The blade and graft image is schematic; it does not demonstrate integration or function in this proposed host.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Revah et al. (2022; updated article with 2026 Author Correction), Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-022-05277-w

Brain2Body canonical glossary, revision 33

Brain2Body programme ethos, current source

Revah et al. (2026), Author Correction: Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-026-10943-4

# Neural Composite

Slide 52 | Reveals 63

An artificial nervous system relates activity to sensing and movement. Measurements can help refine its structure. The pathways are schematic, not measurements.

### Reveal Cues

1. Show the title and conceptual artificial nervous system with activity across sensory, central and motor pathways.

### References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated conceptual illustration of an artificial nervous system

## Physics Engine

Slide 53 | Reveals 64

The proposed physics engine calculates movement under stated mechanical assumptions. This AI-generated image illustrates muscle; it is not a simulation result.

### Reveal Cues

1. Show the title and AI-generated illustration of an ape’s musculature.

### References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated conceptual illustration of an ape with exposed musculature

## World Model

Slide 54 | Reveals 65

The proposed world model aims to simulate hundreds of possible movements over time. The AI-generated illustration suggests parallel movement with motion ghosts; it is not measured simulation data, an exact count or a clinical result.

### Reveal Cues

1. Show the title and AI-generated motion illustration. Distinguish the schematic ghosts from measured simulation or biological results.

### References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated illustration created for this presentation; not measured simulation data.

## Neural Composite

Slide 55 | Reveals 66-68

The proposed neural composite supplies defined inputs. The physics engine calculates mechanical consequences. The world model supplies the task, cues and feedback. Check the interfaces, timing and assumptions, then compare the complete prediction with independent movement. A convincing animation doesn't establish that every hidden component is biologically correct.

### Reveal Cues

1. Reveal Neural Composite with its down arrow; identify provenance and declared neural inputs.

2. Add Physics Engine with its down arrow; explain mechanical units, timing and constraints.

3. Add World Model; explain permitted information, task outcomes and end-to-end independent tests.

### References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

## Large Movement Models

Slide 56 | Reveals 69-70

A virtual arm can follow a path because we gave it the answer. A movement model has to predict the path from declared inputs, then face an independent measurement. Large Synaptic Models names a further ambition; neither name proves that we have mapped a nervous system or trained a successful model.

### Reveal Cues

1. Reveal Large Movement Models with its outgoing down arrow; define a declared movement prediction.

2. Add Large Synaptic Models; distinguish a proposed model name from evidence of synaptic coverage.

### References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

## STAGE 3: Brain2Brain

Slide 57 | Reveals 71

Brain2Brain asks whether information from one nervous system can help another with a stated task. The proposed link and any retained learning require separate tests.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

## 3.1: Mind-Meld

Slide 58 | Reveals 72

Ask whether information from one person can help another with a defined task. A drawn link is only the proposed experiment; compare it with a control.

The direct connection is illustrative; it does not show a demonstrated transfer of experience.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

## 3.2: NeuroTwin

Slide 59 | Reveals 73

NeuroTwin names the proposed biological trainer. The two participants need not be identical, and the pictured role does not establish a working link.

The single animal in the maze is an illustrative trainer view.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

## 3.3: NeuroTransfer

Slide 60 | Reveals 74

Test the proposed information transfer against matched practice or a control link. Help while connected is a different outcome from learning retained afterwards.

The proposed brain-to-brain link runs from the trainer on the left to the trainee on the right. The enlarged cutaways show contacts on simplified brain models; their anatomical fit is unvalidated, and no transfer is demonstrated.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Identify the proposed left-hand trainer as sender and right-hand trainee as recipient; no transfer is demonstrated.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

## 3.4: NeuroTwin

Slide 61 | Reveals 75

The trainer appears again. The outline does not yet say what changes, so a second participant or new condition should not be assumed.

The single animal in the maze is an illustrative trainer view.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

## 3.5: NeuroTransfer

Slide 62 | Reveals 76

Return to NeuroTransfer and test what remains after trainer information stops. Retention and transfer to another task need their own evidence.

The proposed brain-to-brain link runs from the trainer on the left to the trainee on the right. The enlarged cutaways show contacts on simplified brain models; their anatomical fit is unvalidated, and no transfer is demonstrated.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Identify the proposed left-hand trainer as sender and right-hand trainee as recipient; no transfer is demonstrated.

### References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

# STAGE 4: Agent2Brain

Slide 63 | Reveals 77

Agent2Brain asks whether an agent can help a model animal with a stated task through a declared feedback route. Assistance and retained learning are different outcomes.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

## 4.1: Mind-Merge

Slide 64 | Reveals 78

An agent offers information during an animal task. Specify what it observes, how feedback reaches the animal and whether the measured outcome changes.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 4.2: NeuroTrainer

Slide 65 | Reveals 79

The proposed agent predicts or suggests a correction. Test whether its feedback helps beyond a simpler comparison using the same task.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 4.3: NeuroTrainee

Slide 66 | Reveals 80

The model animal performs the task with a stated feedback channel. Measure its action independently; a prediction is not the result.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 4.4: NeuroTrainer

Slide 67 | Reveals 81

Return to the trainer and check what its information actually contributed. The repeated name does not by itself mean there is a second agent.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 4.5: NeuroTrainee

Slide 68 | Reveals 82

Test the trainee after the trainer's help stops. Retained performance and success on a new task need separate measurements.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

# STAGE 5: Agent Vs Neurology

Slide 69 | Reveals 83

Agent Vs Neurology begins with a specified condition, functional problem and measured outcome. It is a proposed comparison, not a demonstrated treatment.

### Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

## 5.1: Pick Your Disease

Slide 70 | Reveals 84

Choose a condition and define a measurable functional problem. Naming a disease in a model does not establish a treatment.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 5.2: Pick Your Agent

Slide 71 | Reveals 85

Choose the agent for a stated prediction, planning or control task. Test what it adds against a simpler comparison.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 5.3: Neural Composite

Slide 72 | Reveals 86

Combine the observations that are actually available, with their timing and uncertainty. A missing recording does not become measured data because a model contains a place for it.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 5.4: Physics Engine

Slide 73 | Reveals 87

The physics engine supplies declared mechanics. Check its movement and timing against independent measurements; a convincing animation is not enough.

### Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

## 5.5: World Model

Slide 74 | Reveals 88

Give the agent a declared world and task, then test on conditions it has not already seen. Success in simulation does not by itself establish a biological or clinical result.

This AI-generated illustration shows parallel movement possibilities; it is not a measured simulation or a biological result.

### Reveal Cues

1. Show the numbered step and AI-generated movement illustration together. Distinguish the illustration from measured simulation and biological results.

### References

AI-generated illustration created for this presentation; not measured simulation data.

## Muscle Function First

Slide 75 | Reveals 89

Start with muscle function because we can compare the predicted action with what the body actually does. Electrical activity, force and movement are different measurements. Organ function adds new

questions and new measurements; a successful hand model would not automatically explain the heart or gut. Each extension has to earn its own evidence.

### Reveal Cues

1. Show the title, image and subtitle together. Use the inside view to explain mechanics, then introduce organ function as a separately measured extension.

### References

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update https://pubmed.ncbi.nlm.nih.gov/25277737/

NICHD, What are the parts of the nervous system? https://www.nichd.nih.gov/health/topics/neuro/conditioninfo/Pages/parts.aspx

## Myoception

Slide 76 | Reveals 90

Myoception is the working label here for sensing related to muscle. Say what is measured and what is inferred: EMG records muscle electrical activity, not every sensory signal or force.

### Reveal Cues

1. Show Myoception and its illustration. Define the working scope and distinguish muscle activity from sensory measurement.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## Viscероception

Slide 77 | Reveals 91

Viscероception is the working label here for sensing related to internal organs. A heartbeat, an associated neural signal and a person's experience are related but different observations.

The front and back views show selected organs, including the stomach and kidneys. They are incomplete illustrations: the positions and fit of structures from different atlases have not been validated.

### Reveal Cues

1. Show Viscероception and its illustration. Give a concrete organ example and distinguish its activity, neural measurements and subjective report.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

## Then We Do It In Humans

Slide 78 | Reveals 92

The human version would need human evidence. An animal result can guide the question; it cannot supply the answer for a person. Whole-Body Emulation is the ambition. Begin with a declared task

and measurement, show where the model works, and keep missing systems visible. A realistic body model alone cannot establish biological function.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the whole-body ambition while keeping the first bounded human task and missing coverage explicit.

### References

Lorach et al. (2023), Walking naturally after spinal cord injury using a brain-spine interface https://pubmed.ncbi.nlm.nih.gov/37225984/

## Disorder

Slide 79 | Reveals 93-94

Now bring the measurements back to the original difficulty. Can they distinguish a useful relationship that a simpler model missed? Recognising a condition, explaining its mechanism and improving it are separate tests. The arrow remains a research direction; neither a diagnostic nor a treatment result is established here.

### Reveal Cues

1. Reintroduce Disorder and its attached arrow as the original observable difficulty.
2. Add Disease; ask what independent new prediction supports a condition-specific explanation.

## Maximally Invasive

Slide 80 | Reveals 95-96

The aim is to move toward approaches that place less burden on the person. Invasiveness, accuracy and everyday intrusiveness are separate questions; each proposed method needs its own comparison and evidence.

### Reveal Cues

1. Reveal Maximally Invasive with its outgoing down arrow; define the proposed starting contrast without prescribing a procedure.
2. Add Minimally Intrusive; distinguish burden from accuracy and require an appropriate comparison.

### References

Brain2Body programme ethos, current source

## Rehabilitation

Slide 81 | Reveals 97-98

Rehabilitation is the work toward better function; restoration is the recovery we hope to measure. Help during an action can be useful. Improvement that remains after trainer guidance ends needs a separate test, with any necessary bridge kept constant. Clinical benefit must be checked through meaningful outcomes, not inferred from a smoother simulation.

### Reveal Cues

1. Show Rehabilitation with its outgoing down arrow.
2. Add Restoration; separate support during action, retained function and clinical benefit.

### References

Biasiucci et al. (2018), Brain-actuated functional electrical stimulation elicits lasting arm motor recovery after stroke https://pubmed.ncbi.nlm.nih.gov/29925890/

## Enablement

Slide 82 | Reveals 99-100

Enablement helps someone complete a task. Enhancement would improve a stated ability beyond a fair comparison. A better score with help does not prove faster learning. Match practice and goals, stop trainer-specific guidance, and test later performance and a new task variation. Keep assistance, learning and transfer as separate results.

### Reveal Cues

1. Introduce Enablement and the arrow to the next proposed ambition.

2. Add Enhancement; define fair comparison, retained performance and task-transfer limits.

## ElectroMagnetism Became Everything

Slide 83 | Reveals 101

Faraday's beginning was modest, not literally nothing. He trained as a bookbinder and gained access to books, people and a laboratory. His experiments helped turn an unseen electrical relationship into something testable. That is the example to borrow: build an instrument, expose the relationship, and let other people check what happens.

### Reveal Cues

1. Show the title, image and subtitle together. Connect induction and later technology to the need for a checkable brain–body instrument, without promising the same historical outcome.

### References

Royal Institution, Michael Faraday (1791–1867) https://www.rigb.org/explore-science/explore/person/michael-faraday-1791-1867

## 2 Papers

Slide 84 | Reveals 102

The two papers set out the mapping and the proposed interface. Their questions need experiments and independent evidence before their conclusions can be established.

### Reveal Cues

1. Show the title, image and subtitle together. Introduce the two static paper previews without treating the overview image as either manuscript or claiming completed biological results.

## NeuroTranslation: A New Paradigm In Brain-Body Mapping

Slide 85 | Reveals 103

NeuroTranslation asks how declared neural and body measurements can constrain one another during a task. Predict something independent, test unused data, and keep the uncertainty visible. A plausible reconstruction is not a unique hidden brain state.

### Reveal Cues

1. Show the paper title and image together. State the mapping question and its evidential limits.

## Brain2Body: The Next Generation In Body-Brain Interfacing

Slide 86 | Reveals 104

Brain2Body asks what an actual information route would need. Recording a signal, driving a virtual body and delivering a biological effect are different operations. Assistance during a task and learning retained afterwards need separate tests.

### Reveal Cues

1. Read the exact paper title; explain the proposed interface question and separate it from achieved function.

### References

Lorach et al. (2023), Walking naturally after spinal cord injury using a brain-spine interface https://pubmed.ncbi.nlm.nih.gov/37225984/

## Starting A New Field

Slide 87 | Reveals 105

We are proposing a field to study internal-body sensing alongside neural connections. The name is a starting point; the measurements and tests have to establish the results.

### Reveal Cues

1. Show the title, image and subtitle together. Introduce the proposal as a research field, not a completed map or established external standard.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Kitmondo LAB, laboratory microscope image, CC BY 2.0; http://www.kitmondo.com/lab-medical-bioscience-equipment https://commons.wikimedia.org/wiki/File:Nikon_Optiphot_Phase_Contrast_Trinocular_Laboratory_Microscope_(15983505825).jpg

## Interoceptomics

Slide 88 | Reveals 106

Interoception concerns information about internal body state; connectomics maps neural connections. Interoceptomics is the proposed study of how those observations relate. A connection alone does not establish function.

The cutaway shows selected muscles and organs. The colouring is illustrative; it is not a complete or measured map of sensory function.

### Reveal Cues

1. Show the title, image and subtitle together. Define both component terms and explain that the proposed field requires measured functional tests.

### References

Brain2Body programme ethos, current source

## With Your Help

Slide 89 | Reveals 107

The useful next step is an answerable experiment: one task, declared measurements, a fair comparison and an independent outcome. Help can mean equipment, expertise or a criticism that improves the test. Any future biological output route needs separate qualification and funding. No new device or budget is announced here.

### Reveal Cues

1. Show the title, image and subtitle together. Make the invitation concrete through experiment, comparison and independent outcome; announce no invented cost.

## Kung Fu THE “EASY” WAY

Slide 90 | Reveals 108

The “easy” way starts smaller: one action, a few declared measurements and a game or task that exposes mistakes. Non-invasive does not mean effortless or already qualified. Compare ordinary movement control with any claimed neural contribution. A working game would be useful; it would not prove an instant skill download.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the bounded non-invasive task and the comparison needed to show that each added measurement helps.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

## Beat Mike Tyson

Slide 91 | Reveals 109

Mike Tyson is the virtual challenge, not a real opponent in an experiment. Read a movement, control an avatar, and measure what the player or agent actually contributes. A game win is a game result. Learning a lasting biological skill would need separate tests after the trainer's help stops. The death line is a joke.

### Reveal Cues

1. Show the title, film poster and subtitle together. Let the subtitle land, then state the virtual task, contributor comparison and non-harm scope.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

# 1. Map Brain2Body

Slide 92 | Reveals 110-114

Map specified brain-related recordings to a measured action and test the prediction under reserved conditions.

Test how a declared action maps between bodies with different proportions and mechanics.

Use the proposed virtual Mike Tyson challenge to test the control route. Survival is a game outcome.

Distinguish success with trainer-specific help from a skill retained afterwards. A virtual result does not copy a person or a career.

Ask whether modelling action with its sensory consequences improves predictions beyond a simpler model, using reserved outcomes.

### Reveal Cues

1. Reveal item 1, Map Brain2Body; define source, target, units and held-out action.
2. Add item 2, Map Body2Body; introduce the separate cross-body mapping problem.
3. Add item 3, Fight & Survive Mike Tyson; introduce the digital challenge.
4. Add item 4, Beat & Become Mike Tyson; distinguish game success from retained learning and identity.
5. Add item 5, Write Motorsensation Is All You Need; state the comparative research question.

# NeuroHandler

Slide 93 | Reveals 115

NeuroHandler starts with one hand action and checks what happened. EEG records scalp potentials; EMG records muscle activity; joined cameras measure movement. EMS—Electric Muscle Stimulation—is the proposed muscle output. First test what the system predicts and controls. A dexterity gain that remains after help ends is another question.

### Reveal Cues

1. Show the title, image and subtitle together. Explain Manual Dexterity with the declared hand action, synchronized left/right capture and independent outcome.

### References

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update https://pubmed.ncbi.nlm.nih.gov/25277737/

OpenSim documentation, How Inverse Kinematics Works https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

# NeuroJuggler

Slide 94 | Reveals 116

Juggling tests whether the hands work together. Release, flight and catch need the same clock, and the object needs its own measurement. Read EEG and EMG, observe movement, and use EMS as the proposed muscle-output route. Compare the joint-task model with a simpler baseline; separate guided performance from coordination that remains afterward.

### Reveal Cues

1. Show the title, image and subtitle together. Explain Multi-Limb Coordination; connect hand-to-hand transfer, object timing and independently measured joint outcomes.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroBoxer

Slide 95 | Reveals 117

NeuroBoxer tests response timing in a virtual challenge. Mike Tyson is the virtual reference; a win does not prove a copied skill.

The participant is shown in a raised boxing pose; the picture does not establish a working interface or copied skill.

### Reveal Cues

1. Show the title, image and subtitle together. Beat Mike Tyson names the virtual challenge, not a real bout.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroBoxer: Read

Slide 96 | Reveals 118

Read the declared EEG, muscle EMG and calibrated movement observations on a shared clock.

### Reveal Cues

1. Read: identify EEG, muscle EMG and calibrated motion observations with a shared clock.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroBoxer: Write

Slide 97 | Reveals 119

Write to the virtual body, ordinary feedback or a separately qualified muscle-output interface; EMS is proposed.

The Write illustration keeps physical arms at the sides, omits sleeves and retains the cap and two cameras. The television shows first-person digital boxing-gloved arms; this does not demonstrate a physical output route.

### Reveal Cues

1. Write: distinguish software/avatar output from an unqualified biological route; EMS remains a proposed muscle-output route requiring separate validation.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroBoxer: React

Slide 98 | Reveals 120

React by measuring the response against the game event and saying who controlled it. Retained learning needs another test.

The React illustration shows raised arms, sleeves and cameras without a cap. It proposes body-directed output; it does not demonstrate an effective intervention.

### Reveal Cues

1. React: measure response timing and outcome, then distinguish assistance from trainer-off retention.

### References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroArcher

Slide 99 | Reveals 121

NeuroArcher tests two hands doing different jobs in a virtual precision challenge. Robin Hood is illustrative; no person or animal is a target.

The participant is shown with a bow and arrow; the image does not establish a working interface or transferred skill.

### Reveal Cues

1. Show the title, image and subtitle together. Shoot Robin Hood is an illustrative precision challenge, with no actual person or animal targeted.

### References

OpenSim documentation, How Inverse Kinematics Works https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroArcher: Read

Slide 100 | Reveals 122

Read the declared neural, muscle, movement and object observations with the release event on the same clock.

### Reveal Cues

1. Read: align both limbs, object state and release with the declared observations.

### References

OpenSim documentation, How Inverse Kinematics Works https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroArcher: Write

Slide 101 | Reveals 123

Write to the virtual archer or ordinary feedback; EMS is a proposed output, not proof of physical skill transfer.

The Write illustration keeps physical arms at the sides, omits sleeves and retains the cap and two cameras. The television shows first-person digital arms holding a bow; this does not demonstrate a physical output route.

### Reveal Cues

1. Write: name virtual output or ordinary feedback; do not equate EMS with a validated biological writer.

### References

OpenSim documentation, How Inverse Kinematics Works https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## NeuroArcher: React

Slide 102 | Reveals 124

React by measuring release timing and target error. Separate agent assistance from retained learning.

The React illustration shows raised arms, sleeves and cameras without a cap. It proposes body-directed output; it does not demonstrate an effective intervention.

### Reveal Cues

1. React: measure release timing and target error; separate agent assistance and retained human learning.

### References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

## Coming Eventually

Slide 103 | Reveals 125

Actions have sensory consequences. The proposed paper asks whether modelling them together improves a declared task beyond a simpler model. “All You Need” is the hypothesis to challenge, not permission to ignore vision, cognition or context. Name the measured senses, keep the targets out of the inputs, and report where the comparison fails.

The screenshot is from the earlier Attention Is All You Need paper, not the proposed Motorsensation paper.

### Reveal Cues

1. Show the title, image and subtitle together. Identify the Attention Is All You Need screenshot as a historical reference, then state the proposed action/sensory comparison.

### References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

## System 1: FAST

Slide 104 | Reveals 126-128

FAST is the proposed action route. SLOW checks errors and develops changes. COMPRESSED would preserve useful work in a form that is faster or cheaper to use, if the original quality tests still pass. This is a computational and human–machine analogy. It is not a discovered three-part brain architecture or a measured speed gain.

### Reveal Cues

1. Reveal System 1: FAST with the down arrow; introduce the frozen-inference analogy.

2. Add System 2: SLOW with its arrow; explain independent checking and candidate updates.

3. Add System 3: COMPRESSED; explain a proposed accepted compression/partnership, without a biological or performance claim.

### References

Daniel Kahneman (2011), Thinking, Fast and Slow, Microsoft Research lecture
https://www.microsoft.com/en-us/research/video/thinking-fast-and-slow/

## Most Valuable Human Skill

Slide 105 | Reveals 129

The claim is about learning: meeting a new task and keeping what helps after the trainer stops. “The only human skill” is rhetorical emphasis, not a scientific ranking. Test a defined skill with matched practice, delayed retention and an unused variation. A tool that does the difficult part for you may be useful without teaching it. The evolution image is symbolic, not a straight-line account of human evolution.

### Reveal Cues

1. Show the title, image and subtitle together. Explain that the walking-evolution image is symbolic, then distinguish assistance from measured retention.

### References

Smithsonian Human Origins Program, Frequently Asked Questions https://humanorigins.si.edu/education/frequently-asked-questions

## Months & Years

Slide 106 | Reveals 130

Months and years versus weeks and days is an ambition, not a demonstrated speedup. Define the skill and test how long it takes to reach and retain a measured level of performance.

### Reveal Cues

1. Show the title, calendar-and-stopwatch image and subtitle together. Present the time contrast as an ambition and name the matched retention test needed to measure it.

### References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Eric Rothermel, monthly schedule image, Commons CC0
https://commons.wikimedia.org/wiki/File:Monthly_schedule_(Unsplash).jpg

Liilia Moroz, stopwatch image, Commons CC0 https://commons.wikimedia.org/wiki/File:Stopwatch_made_in_USSR.jpg

## Whole-Body Emulation

Slide 107 | Reveals 131

The brain acts through a body in an environment, so the body gives us measurements and consequences to test. A lifelike body does not supply missing neural function. Emulation needs a declared task, accuracy, units, uncertainty and held-out evidence. Whole-body and whole-brain function remain ambitions beyond the bounded relationships proposed here.

### Reveal Cues

1. Show the title, image and subtitle together. Reconnect whole-body and whole-brain emulation; reconnect the five measurement/modelling roles and distinguish validated domain from total functional coverage.

### References

NIST Technical Note 1297, Appendix D.1 Terminology https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-appendix-d1-terminology

# Vertical Column Central Axis

Slide 108 | Reveals 132

Moving your body and regulating its organs use overlapping brain circuits. Somatic function covers body sensation and skeletal movement. Autonomic pathways help regulate organs through sympathetic and parasympathetic outputs. Sensory pathways carry information back from the body.

The shades show illustrative associations with sensation, movement and internal regulation; unshaded regions can also participate. The cortex, thalamus, cerebellum, hypothalamus and brainstem contribute different parts of that work. Breathing is a useful example: a skeletal muscle keeps working automatically, while you can also change the breath deliberately.

This is a constructed medial view of the source-left hemisphere, front to the right, rolled 12° for display. Twenty occupied atlas labels have no model geometry. Some important structures are hidden, and the broad shapes do not resolve every small nucleus. The backslash is a visual shorthand; the supporting outline gives the fuller map and its evidence.

## Reveal Cues

1. Show the existing title, revised image and subtitle together. Identify somatic, autonomic/visceral and shared associations, then relate them through breathing; keep the single-hemisphere visibility limit explicit.

## References


Ding et al. (2016), Comprehensive cellular-resolution atlas of the adult human brain https://pmc.ncbi.nlm.nih.gov/articles/PMC5054943/

Ding et al. (2020), Allen Human Reference Atlas — 3D, v1.0.0 (CC BY 4.0) https://download.alleninstitute.org/informatics-archive/allen_human_reference_atlas_3d_2020/version_1/

Allen Human Reference Atlas 3D v2 https://alleninstitute.github.io/CCF-MAP/descriptions/human_ccf.html

Glasser et al. (2016), A multi-modal parcellation of human cerebral cortex https://www.nature.com/articles/nature18933

Gordon et al. (2023), A somato-cognitive action network alternates with effector regions in motor cortex https://pmc.ncbi.nlm.nih.gov/articles/PMC10172144/

Dum et al. (2016), Motor, cognitive, and affective areas of the cerebral cortex influence the adrenal medulla https://pubmed.ncbi.nlm.nih.gov/27528671/

Ran et al. (2022), A brainstem map for visceral sensations https://www.nature.com/articles/s41586-022-05139-5

Veerakumar et al. (2022), Molecularly defined circuits for cardiovascular and cardiopulmonary control https://www.nature.com/articles/s41586-022-04760-8

Cui et al. (2016), Defining preBötzinger Complex Rhythm- and Pattern-Generating Neural Microcircuits In Vivo https://pmc.ncbi.nlm.nih.gov/articles/PMC4978183/

Keller et al. (2018), Voluntary urination control by brainstem neurons that relax the urethral sphincter https://pmc.ncbi.nlm.nih.gov/articles/PMC6119086/

Allison et al. (1996), Localization of functional regions of human mesial cortex by somatosensory evoked potential recording and by cortical stimulation https://pubmed.ncbi.nlm.nih.gov/8617151/

Sanchez-Larsen et al. (2021), Characterization of the Insular Role in Cardiac Function through Intracranial Electrical Stimulation of the Human Insula https://pubmed.ncbi.nlm.nih.gov/33783847/

Green et al. (2005), Deep brain stimulation can regulate arterial blood pressure in awake humans https://pubmed.ncbi.nlm.nih.gov/16237319/

Lenz et al. (1988), Single-unit analysis of the human ventral thalamic nuclear group: somatosensory responses https://pubmed.ncbi.nlm.nih.gov/3351564/

Dutschmann and Herbert (2006), The Kölliker-Fuse nucleus gates the postinspiratory phase of the respiratory cycle to control inspiratory off-switch and upper airway resistance in rat https://pubmed.ncbi.nlm.nih.gov/16930433/

Short-term training-dependent plasticity of the corticospinal diaphragm control in normal humans https://pubmed.ncbi.nlm.nih.gov/17964864/

Gai and Blessing (1996), Human brainstem preganglionic parasympathetic neurons localized by markers for nitric oxide synthesis https://pubmed.ncbi.nlm.nih.gov/8813279/

## Blood Pressure Monitor

Slide 109 | Reveals 133

First, make a useful nervous-system measurement and test it against an independent reference. Then study a separately qualified way to affect a declared process. The monitor and pacemaker are proposed analogies, not demonstrated devices or treatments.

### Reveal Cues

1. Show the title, image and subtitle together. Explain the proposed measurement first and the separately qualified intervention next; claim no demonstrated device.

### References

Brain2Body programme ethos, current source

## That's How I Know Kung Fu

Slide 110 | Reveals 134

The film gives us the finished download. The experiment has to earn every connection: record, predict, check the action, identify the feedback route, then test what remains after trainer-specific help ends. There is no kung fu to download yet. The invitation is to help turn one missing relationship into a result we can independently check.

### Reveal Cues

1. Show the title, image and subtitle together. Let the closing image land; distinguish the invitation from a guaranteed skill download and finish on the next checkable result.

## eliminatenoise.org

Slide 111 | Reveals 135

Find the project at eliminatenoise.org.

### Reveal Cues

1. Show eliminatenoise.org alone, centered in regular type, as the final slide.